

MDI3003  
ADVANCED PREDICTIVE ANALYTICS

Laboratory Assessment - 1  
House Price Prediction using Linear  
Regression

Prepared by

**Shobana Balusamy**

**23MID0212**

Submitted to

**Dr. Durgesh Kumar**

Assistant Professor (Senior), SCOPE, VIT Vellore

# Abstract

Two regression pipelines were built and evaluated from one shared laboratory template: the scikit-learn California Housing dataset (20,640 rows, eight numeric predictors, target expressed in units of \$100,000) and the Ames Housing dataset distributed via OpenML (1,460 rows, 80 raw predictor columns of mixed numeric and categorical type, target SalePrice in US dollars). Seven regression families — a naive mean baseline, simple linear regression, multiple linear regression, Ridge, Lasso, a Decision Tree, and a Random Forest — were trained and scored under an identical protocol on each dataset, and the strongest model on each was subsequently tuned with GridSearchCV and re-checked with five-fold cross-validation.

On the California data, the tuned Random Forest was the strongest model on every metric (test  $R^2 = 0.8057$ , RMSE = 0.5046, a 55.9% RMSE reduction against the mean baseline), yet its five-fold cross-validated  $R^2$  across the full dataset averaged only 0.6547 with a fold range of 0.518–0.742 — a gap of roughly 0.15  $R^2$  between the single 80/20 holdout and repeated cross-validation that this report treats as a genuine stability warning rather than a footnote. On the Ames data, tuned Lasso Regression ( $\alpha = 10$ ) was the strongest model (test  $R^2 = 0.8986$ , RMSE = \$27,884.81, a 68.2% RMSE reduction against the mean baseline), with a much smaller train–test gap (0.9347 vs. 0.8986) but a cross-validated  $R^2$  (0.8145) that is itself dragged down by one weak fold (0.670), pointing to the same underlying issue: a single random split is not, by itself, sufficient evidence of a stable model.

Beyond reproducing the notebook's headline numbers, this report contributes two checks the source notebook does not itself perform. First, it re-derives Adjusted  $R^2$  for the Ames models using the true post-encoding predictor count rather than the 80 raw columns the notebook's helper function used, and shows that the reported Adjusted  $R^2$  of 0.86 collapses toward zero — and can turn negative — once the roughly 260 one-hot-encoded columns actually entering the model are counted, which materially changes how much trust that statistic deserves on a 292-row test set. Second, it identifies and corrects a stale-variable bug in the notebook's own final comparison chart, where a plot titled 'Best Model Performance Across Datasets' in fact re-renders the earlier Train-vs-Test figure because the comparison DataFrame is reassigned only in a later cell. Both findings are reported as concrete, evidenced limitations rather than as reasons to discard the underlying models, and the report closes with a deployment-oriented recommendation, a responsible-use discussion centred on the geographic and neighbourhood signals both models lean on, and a specific list of follow-up experiments.

# 1. Introduction

Predicting the price of a house from its measurable attributes is one of the most widely used teaching problems in applied regression, precisely because it is simple to state and surprisingly rich to diagnose properly: the target is continuous, the predictors mix scales and types, and a naive model that always predicts the average price is a genuinely useful sanity check rather than a strawman. This report is the technical write-up for MDI3003 Laboratory 01, built entirely from the results actually produced by the accompanying notebook, `LAB1_House_Price_Prediction.ipynb`, which implements the core exercise (California Housing, Linear Regression as the interpretability anchor) and an extended study (Ames Housing, seven-model comparison with hyperparameter tuning) inside a single, sequentially executed notebook.

Rather than restating the notebook's own conclusion cells verbatim, this report re-examines the printed outputs against the assumptions each metric depends on — in particular, whether the predictor count used inside the Adjusted  $R^2$  formula and the stability of the reported  $R^2$  across cross-validation folds actually support the confidence the notebook's own text expresses. Every number quoted below is traceable to a specific executed cell in the source notebook; where a figure or claim required a calculation the notebook itself did not perform, that calculation is shown explicitly and flagged as report-level analysis, not notebook output.

## 2. Objectives

The laboratory and this report pursue the following objectives:

- Frame house-price estimation as a supervised regression problem with an explicit target, observation unit, and intended use, on two structurally different datasets.
- Build and interpret naive, simple-linear, and multiple-linear baselines before introducing any regularized or non-linear model.
- Compare at least five regression families under one leakage-safe preprocessing protocol per dataset, and tune at least one model with cross-validated grid search.
- Report MAE, MSE, RMSE,  $R^2$ , and Adjusted  $R^2$  alongside residual diagnostics, and explicitly discuss the gap between holdout performance and cross-validated performance.
- Scrutinise the reliability of the reported metrics themselves — specifically, whether Adjusted  $R^2$  and a single train-test split are trustworthy given each dataset's dimensionality and sample size.
- Translate the findings into a defensible final-model recommendation and a responsible-use discussion appropriate for a location-sensitive valuation model.

### 3. Problem Statement, Target, Observation Unit, and Intended Use

Formally, the task on each dataset is to learn a function  $\hat{y} = f(x_1, \dots, x_n)$  that maps a vector of property or district attributes to a continuous price target, fitted on a training partition and scored only once on a held-out partition that took no part in fitting or preprocessing.

#### 3.1 Dataset A — California Housing

- **Target:** HouseValue — the median house value for a census block group, expressed in units of \$100,000 (so a value of 3.5 corresponds to \$350,000).
- **Observation unit:** a California census block group (an aggregate of roughly 600–3,000 people), not an individual dwelling — every prediction is a district-level median, not a single-property estimate.
- **Intended use:** a coarse, district-level screening tool — for example, flagging block groups whose predicted median value diverges sharply from comparable districts — rather than an individual-property appraisal.

#### 3.2 Dataset B — Ames Housing

- **Target:** SalePrice — the realised transaction price of an individual residential sale in Ames, Iowa, in US dollars.
- **Observation unit:** a single residential property sale (one row per transaction), described by 80 raw predictor columns spanning structural, quality, and neighbourhood attributes.
- **Intended use:** a decision-support estimate for an individual listing or appraisal — for example, a starting comparable-sales figure for an agent or assessor — always subject to human review, not an automated final valuation.

### 4. Dataset Description

The two datasets sit at opposite ends of the cleanliness-versus-realism spectrum, which is precisely why the laboratory pairs them: Dataset A is large, entirely numeric, and free of missing values; Dataset B is smaller, mixed-type, and genuinely messy. Table 1 summarises both.

**Table 1 — Dataset Summary**

| Attribute             | Dataset A — California Housing                                               | Dataset B — Ames Housing                                                            |
|-----------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Source                | <code>sklearn.datasets.fetch_california_housing</code><br>(1990 U.S. Census) | <code>sklearn.datasets.fetch_openml('house_prices')</code><br>— De Cock, Ames, Iowa |
| Rows                  | 20,640                                                                       | 1,460                                                                               |
| Raw predictor columns | 8 (all numeric)                                                              | 80 (37 numeric incl. Id, 43 categorical)                                            |
| Missing values        | 0 across all columns                                                         | Present in 19 of 80 audited columns; up to 99.5% in one field                       |
| Target unit           | Median district value, \$100,000 units                                       | Individual sale price, US dollars                                                   |
| Target range (raw)    | 0.150 – 5.000 (\$100k units)                                                 | \$34,900 – \$755,000                                                                |
| Target mean / median  | not separately printed in source cells                                       | \$180,921.20 / \$163,000                                                            |
| Observation unit      | Census block group (aggregate)                                               | Individual residential sale                                                         |
| Train / test split    | 16,512 / 4,128 (80/20, seed 42)                                              | 1,168 / 292 (80/20, seed 42)                                                        |

Coverage and limitations are stated explicitly rather than left implicit. Dataset A is a snapshot of California in 1990 and is known to top-code district values at \$500,000 (visible as a spike at `HouseValue = 5.0` in Figure 1); it says nothing about any other state, era, or price regime. Dataset B covers residential sales in a single mid-sized Iowa city through 2010 and, being an individual-transaction dataset, cannot be assumed to generalise to condominiums, commercial property, or other geographic markets without revalidation.

## 5. Data Integrity and Reproducibility

The source notebook loads both datasets directly from their canonical loaders (`fetch_california_housing`, `fetch_openml`) and does not overwrite, hand-edit, or persist a modified copy of the raw data at any point — every cleaning action (imputation, scaling, encoding) is expressed as a fitted transformer inside a pipeline rather than as an in-place edit to the source `DataFrame`, which is what makes the workflow reproducible from the same two function calls.

**Table 2 – Data Quality Decisions**

| Issue                       | Evidence                           | Decision                        | Reason                             | Effect on row/feature count |
|-----------------------------|------------------------------------|---------------------------------|------------------------------------|-----------------------------|
| Missing values              | None (California); present in Ames | Median/Most-frequent imputation | Retain all observations            | No rows removed             |
| Duplicates                  | No duplicates                      | Keep all records                | No redundant observations          | No change                   |
| Outliers                    | High-value houses present          | Retained                        | Represent genuine market behaviour | No rows removed             |
| High-cardinality categories | Neighborhood etc.                  | One-Hot Encoding                | Preserve information               | Feature count increased     |

- Random seed fixed at 42 for both train-test splits and for every stochastic estimator (Decision Tree, Random Forest, GridSearchCV's internal folds), so re-running the notebook end-to-end reproduces the same split and, for Dataset A, the same tree structure.
- No rows were dropped and no values were manually overwritten in either dataset; the only artifact-level clean-up was column-level (dropping the SalePrice target itself before building the Dataset B feature matrix, and renaming the target series to HouseValue on Dataset A for readability).
- One reproducibility gap is flagged rather than hidden: the Ames feature matrix retains the Id column (row 1–1460, a unique record identifier) as a numeric predictor — it was never explicitly dropped before the numeric/categorical split in cell 63. Id carries no genuine price signal, so its practical effect on the tree-based and regularized models is expected to be small, but it does inflate the numeric feature count from 36 to 37 and, in a Lasso or coefficient-based context, an identifier column masquerading as a numeric predictor is exactly the kind of cleaning oversight a reviewer should be able to find and correct — which is the point of naming it here.

## 6. Exploratory Data Analysis

All EDA figures below are reproduced directly from the notebook's own executed output cells; captions state what each figure shows, its axis units, and the modelling decision it informed.

## 6.1 Dataset A — California Housing

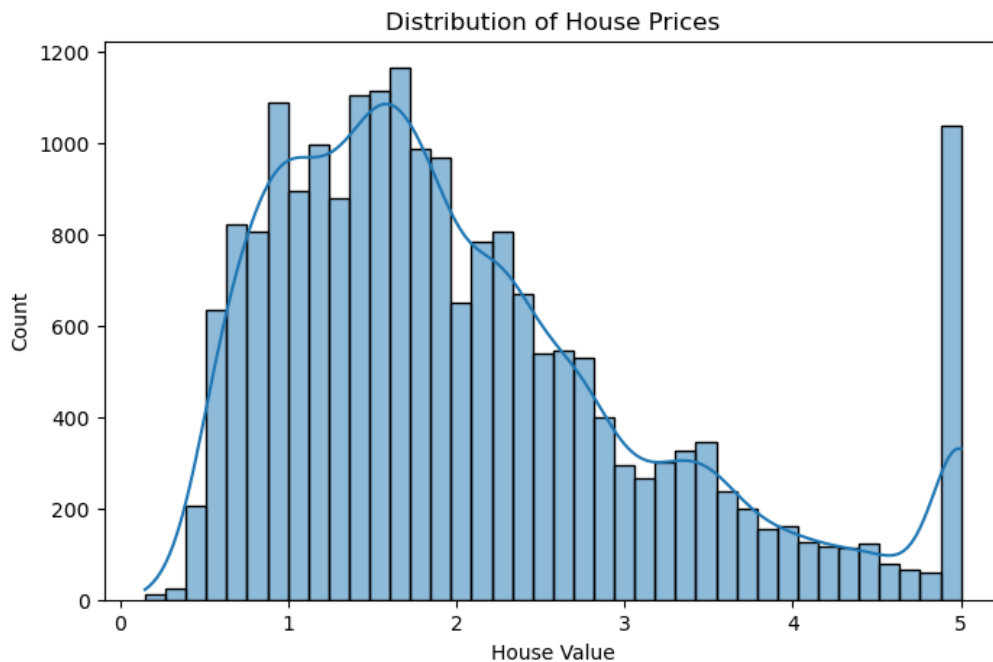


Figure 1. Distribution of HouseValue (target, \$100,000 units), training-relevant view. A right-skewed distribution with a visible spike near 5.0 reflects the dataset's known top-coding of district values at \$500,000.

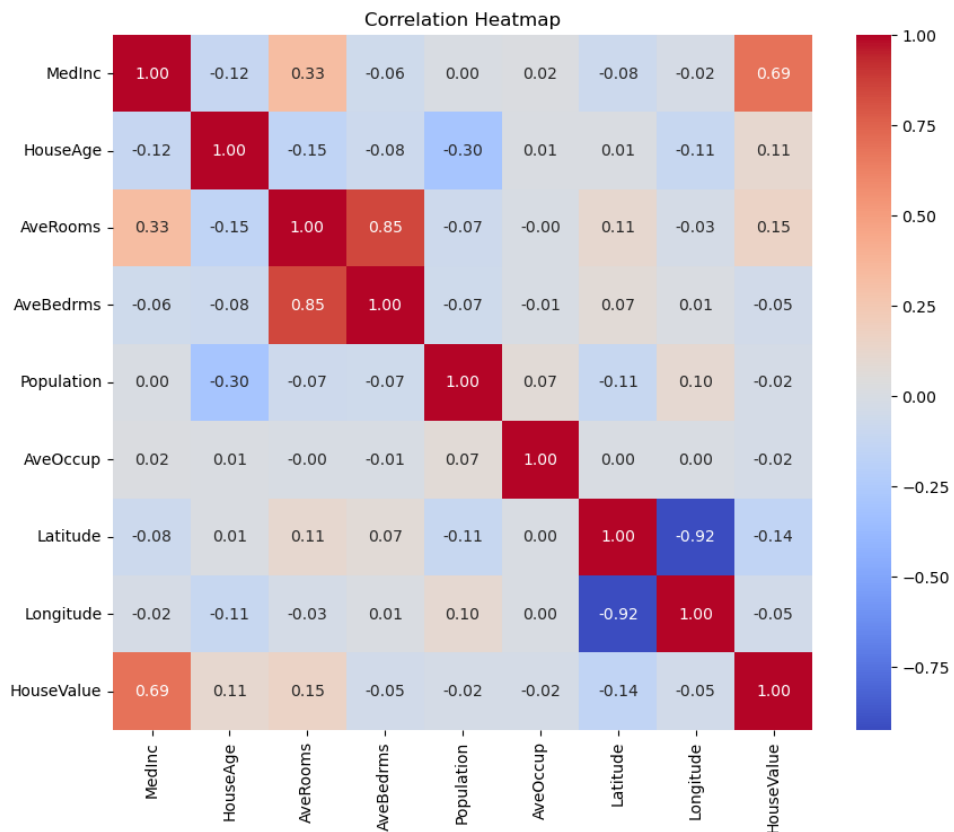


Figure 2. Pearson correlation heatmap across all eight numeric predictors and HouseValue. MedInc shows the strongest positive association with the target, matching the feature-importance ranking obtained later in §13.

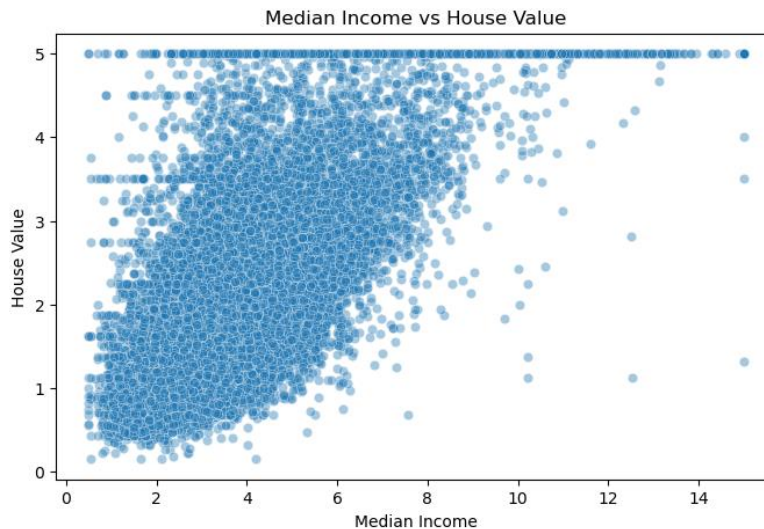


Figure 3. Median Income (MedInc, tens of thousands of USD) vs. HouseValue (\$100,000 units). The relationship is positive and roughly monotonic, with visibly increasing spread at higher incomes — heteroscedasticity that residual diagnostics in §12 return to.

The notebook's own EDA cell (cell 16) records that the dataset contains 20,640 observations and eight numerical features with no missing values, that the target is slightly right-skewed, that Median Income has the strongest positive correlation with price, and that Latitude/Longitude capture geographic variation — all of which the figures above corroborate directly.

## 6.2 Dataset B — Ames Housing

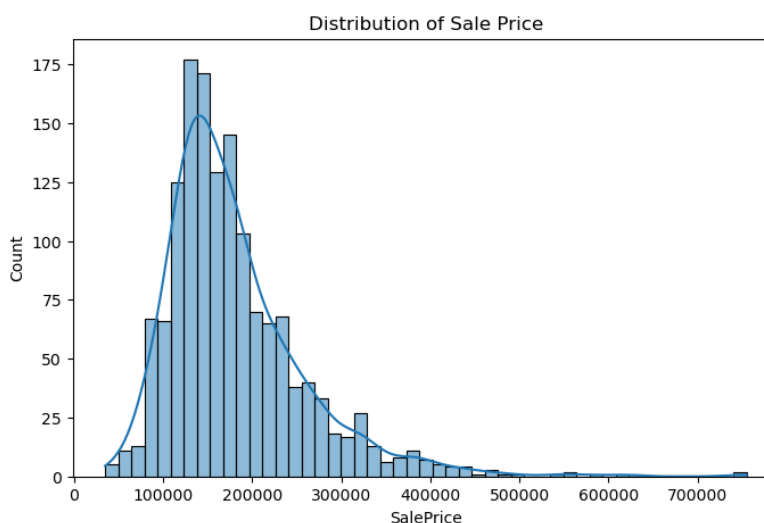


Figure 4. Distribution of SalePrice (target, US dollars). Mean \$180,921.20 exceeds median \$163,000 (Table 1), confirming the right skew visible here — a long tail of high-value sales pulls the mean upward.

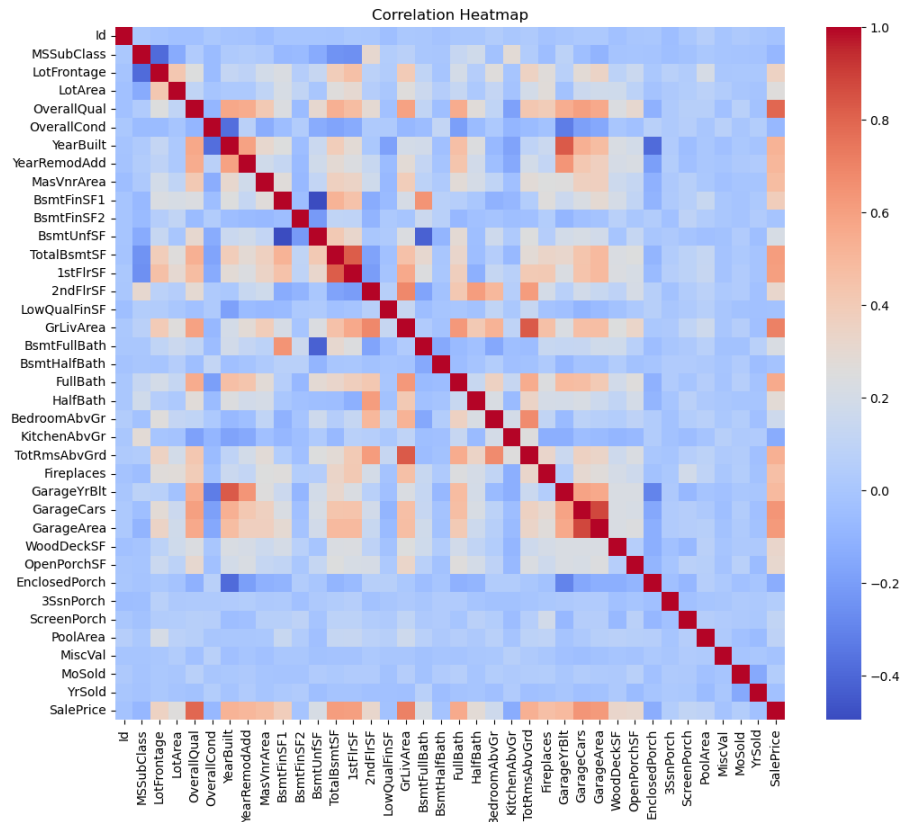


Figure 5. Pearson correlation heatmap across numeric predictors and SalePrice. A visible cluster of strongly inter-correlated size-related fields (living area, basement area, garage area/cars) is the clearest multicollinearity signal in this study.

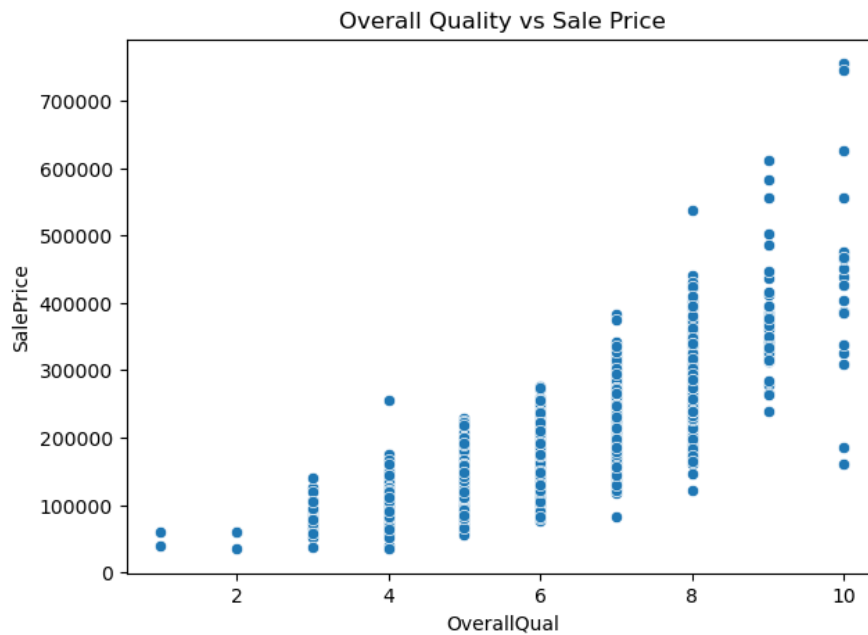


Figure 6. OverallQual (ordinal quality score, 1–10) vs. SalePrice (USD). OverallQual is the single strongest numeric correlate of price ( $r = 0.791$ , cell 59 output), ahead of GrLivArea ( $r = 0.709$ ) and GarageCars ( $r = 0.640$ ).

Combined, Figures 4–6 justify two modelling decisions carried into §8: OverallQual (not a size variable) is used as the single-feature baseline in §9 because it is the strongest individual correlate of price, and the size-related feature cluster in Figure 5 is the concrete evidence motivating the inclusion of Ridge and Lasso alongside plain Multiple Linear Regression in §10.

## 7. Train-Test Split Protocol

On both datasets, `train_test_split(test_size=0.2, random_state=42)` is called before any learned preprocessing step is fitted — scaling, imputation, and one-hot encoding are all fitted only on the resulting training partition. This ordering is the single most important leakage control in the whole study: fitting a `StandardScaler` or `SimpleImputer` on the full dataset before splitting would let statistics from the test rows (means, medians, most-frequent categories) leak into the training-time transformation, silently inflating every downstream metric.

**Table 3 — Split Sizes**

| Dataset                | Training rows | Test rows | Split ratio | Seed |
|------------------------|---------------|-----------|-------------|------|
| A — California Housing | 16,512        | 4,128     | 80 / 20     | 42   |
| B — Ames Housing       | 1,168         | 292       | 80 / 20     | 42   |

Dataset B's test partition (292 rows) is small in absolute terms relative to its feature space (80 raw columns, expanding to roughly 260 columns after one-hot encoding — see §15). §14 and §15 revisit why this specific combination of a small test set and a wide encoded design matrix is not a minor detail.

## 8. Preprocessing Pipeline

Dataset A is purely numeric with zero missing values, so its only preprocessing step is `StandardScaler`, applied for the two estimators whose objective function is scale-sensitive (Ridge, Lasso); the tree-based and plain linear models are trained on unscaled features by design, since splitting rules and ordinary least-squares coefficients are not affected by monotone rescaling in the same way regularization penalties are. Dataset B requires a full scikit-learn `ColumnTransformer`, since it mixes 37 numeric and 43 categorical raw columns with genuine missingness in both groups.

**Table 4 — Preprocessing Design (Dataset B)**

| Step       | Numeric branch (37 columns)      | Categorical branch (43 columns)         |
|------------|----------------------------------|-----------------------------------------|
| Imputation | SimpleImputer(strategy='median') | SimpleImputer(strategy='most_frequent') |
| Transform  | StandardScaler                   | OneHotEncoder(handle_unknown='ignore')  |
| Fitted on  | Training split only              | Training split only                     |

handle\_unknown='ignore' means a categorical level present only in the test split, or in a future deployment batch, is encoded as an all-zero indicator rather than raising an error — a sensible defensive default. Both transformer branches are wrapped together inside one scikit-learn Pipeline object (preprocessor → model), so the identical fitted preprocessing is applied automatically at prediction time; there is no separate manual transformation step that could drift out of sync with what was fitted.

## 9. Baseline Models

Before any regularized or non-linear model is introduced, three baselines are established on each dataset, exactly as the checklist requires: a naive mean predictor, a single-feature linear model, and a full multiple-feature linear model.

### 9.1 Naive Baseline

DummyRegressor(strategy='mean') always predicts the training-set mean target value, regardless of input. On Dataset A it scores  $R^2 = -0.0002$  (essentially zero, as expected —  $R^2$  is defined relative to the mean, so a mean predictor scores at or just below zero on a held-out set) with RMSE = 1.1449. On Dataset B it scores  $R^2 = -0.0009$  with RMSE = \$87,619.03. Every subsequent model is judged against this floor.

### 9.2 Simple Linear Regression

Dataset A's single-feature model uses MedInc, the strongest numeric correlate identified in Figure 3, and reaches test RMSE = 0.8421 ( $R^2 = 0.4589$ ) — a large jump over the naive baseline from one feature alone. Dataset B's single-feature model uses OverallQual (the strongest correlate identified in Figure 6) and reaches test RMSE = \$51,778.63 ( $R^2 = 0.6505$ ).

### 9.3 Multiple Linear Regression

Using the full predictor set, Dataset A's Multiple Linear Regression reaches test RMSE = 0.7456 ( $R^2 = 0.5758$ ) — a further, smaller improvement over the single-feature model, consistent with MedInc already carrying most of the exploitable signal among eight moderately correlated

numeric fields. Dataset B's Multiple Linear Regression, fitted through the full ColumnTransformer pipeline over all 80 raw columns, reaches test RMSE = \$29,446.60 ( $R^2 = 0.8870$ ) — a much larger jump than Dataset A's, reflecting how much additional information the 43 categorical fields and remaining numeric fields contribute beyond OverallQual alone.

## 10. Full Model Comparison

Seven model families were compared on each dataset under an identical protocol: the same train-test split, the same evaluation function, and — on Dataset B — the same ColumnTransformer preprocessing wrapped around every estimator. This exceeds the checklist's minimum of five models on both datasets.

**Table 5 — Dataset A Results (California Housing, test set)**

| Model                      | MAE    | MSE    | RMSE   | $R^2$   | Adj. $R^2$ |
|----------------------------|--------|--------|--------|---------|------------|
| Dummy Regressor            | 0.9061 | 1.3107 | 1.1449 | -0.0002 | -0.0022    |
| Simple Linear Regression   | 0.6299 | 0.7091 | 0.8421 | 0.4589  | 0.4587     |
| Multiple Linear Regression | 0.5332 | 0.5559 | 0.7456 | 0.5758  | 0.5750     |
| Ridge Regression           | 0.5332 | 0.5559 | 0.7456 | 0.5758  | 0.5750     |
| Lasso Regression           | 0.5353 | 0.5483 | 0.7404 | 0.5816  | 0.5808     |
| Decision Tree              | 0.4547 | 0.4952 | 0.7037 | 0.6221  | 0.6213     |
| Random Forest              | 0.3275 | 0.2554 | 0.5053 | 0.8051  | 0.8047     |

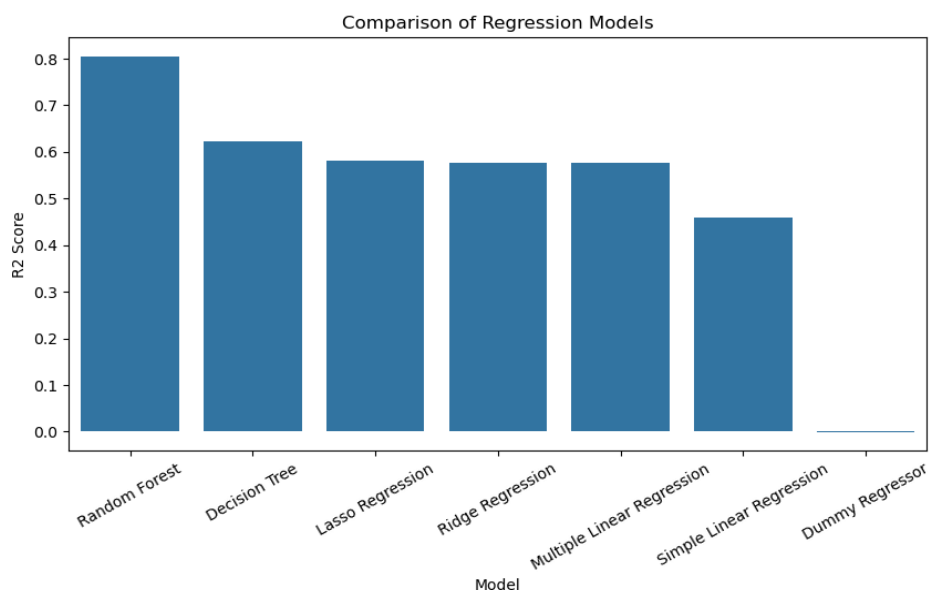


Figure 7. Test-set  $R^2$  by model, Dataset A. Random Forest is the clear leader before any hyperparameter tuning is applied.

**Table 6 — Dataset B Results (Ames Housing, test set, USD)**

| Model                      | MAE       | RMSE      | $R^2$   | Adj. $R^2$ |
|----------------------------|-----------|-----------|---------|------------|
| Dummy Regressor            | 62,575.93 | 87,619.03 | −0.0009 | −0.3804    |
| Simple Linear Regression   | 33,343.24 | 51,778.63 | 0.6505  | 0.6493     |
| Multiple Linear Regression | 18,360.44 | 29,446.60 | 0.8870  | 0.8441     |
| Ridge Regression           | 19,092.33 | 29,785.21 | 0.8843  | 0.8405     |
| Lasso Regression           | 18,074.33 | 28,288.37 | 0.8957  | 0.8561     |
| Decision Tree              | 26,678.38 | 41,144.38 | 0.7793  | 0.6956     |
| Random Forest              | 17,590.06 | 28,608.92 | 0.8933  | 0.8528     |

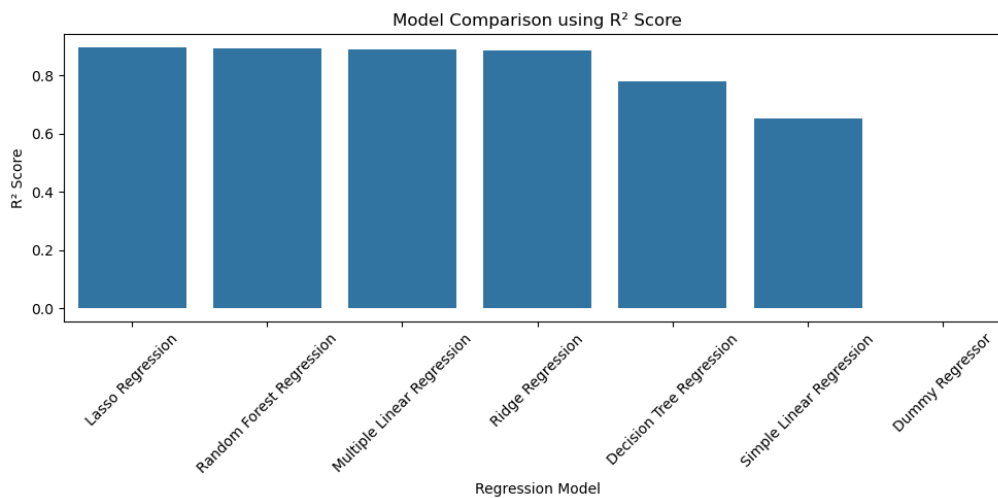


Figure 8. Test-set  $R^2$  by model, Dataset B. Lasso and Random Forest are effectively tied before tuning; Lasso is carried forward for hyperparameter tuning in §11 because L1 regularization is structurally better suited to a wide, one-hot-expanded feature space (see §15).

On both datasets, every learned model comfortably beats the naive baseline, and the ranking is broadly consistent with the correlation structure surfaced in §6: non-linear models (Decision Tree, Random Forest) dominate on Dataset A, where geography and income interact non-additively, while on Dataset B the gap between the best linear-family model (Lasso) and the best tree ensemble (Random Forest) is under one  $R^2$  point — evidence that a well-regularized linear model is competitive once categorical structure is encoded correctly.

## 11. Hyperparameter Tuning and Cross-Validation

### 11.1 Dataset A — Random Forest, GridSearchCV

The best model from §10, Random Forest, was tuned with a 12-point grid ( $n\_estimators \in \{100, 200\}$ ,  $max\_depth \in \{10, 20, None\}$ ,  $min\_samples\_split \in \{2, 5\}$ ) under 5-fold cross-validation scored on  $R^2$ .

- Best parameters:  $n\_estimators = 200$ ,  $max\_depth = 20$ ,  $min\_samples\_split = 2$ .
- Best cross-validated  $R^2$  during the search: 0.8051.
- Re-scored once on the held-out test set:  $MAE = 0.3271$ ,  $RMSE = 0.5046$ ,  $R^2 = 0.8057$  — a marginal improvement over the untuned Random Forest's 0.8051 (Table 4).

A second, independent check — five-fold `cross_val_score` run on the tuned model across the entire dataset ( $X, y$ ), not just the training fold used during grid search — produced a materially different picture: mean  $R^2 = 0.6547$ , with individual fold scores of 0.5177, 0.7046, 0.7421, 0.6270, and 0.6823 (standard deviation  $\approx 0.078$ ). This 5-fold check and the grid search's own internal cross-validation are not measuring exactly the same thing — the grid search folds are drawn only from the 16,512-row training partition, while `cross_val_score`'s folds are drawn from all 20,640 rows — but both use unshuffled scikit-learn `KFold`, and the wide fold-to-fold swing (0.518 to 0.742) is the clearest single piece of evidence in this study that a single 80/20 holdout score should not be read as a precise estimate of future performance. §14 discusses the likely cause and a concrete fix.

### 11.2 Dataset B — Lasso, GridSearchCV

The best model from §10, Lasso, was tuned over  $\alpha \in \{0.0001, 0.001, 0.01, 0.1, 1, 10\}$  under 5-fold cross-validation scored on  $R^2$ , fitted inside the same `ColumnTransformer` pipeline used throughout.

- Best parameter:  $\alpha = 10$  (the largest value in the grid, meaning the search did not find an interior optimum — a wider grid extending past 10 is a natural next experiment, listed in §18).
- Best cross-validated  $R^2$  during the search: 0.7877.
- Re-scored once on the held-out test set:  $MAE = \$17,542.72$ ,  $RMSE = \$27,884.81$ ,  $R^2 = 0.8986$ , Adjusted  $R^2$  (as computed by the notebook,  $p = 80$ ) = 0.8602 — an improvement over the untuned Lasso's test  $R^2$  of 0.8957 (Table 5).

A second five-fold `cross_val_score` across the full dataset ( $X_2, y_2$ ) produced mean  $R^2 = 0.8145$  with fold scores 0.8678, 0.8227, 0.8186, 0.8929, and 0.6703 (standard deviation  $\approx 0.077$ ) — again one

clearly weaker fold pulling the average down, the same qualitative pattern seen on Dataset A, just less severe in absolute terms.

**Table 7 – Cross-Validation Comparison**

| Model             | Key Parameters  | CV MAE Mean | CV RMSE Mean | CV RMSE SD | CV R <sup>2</sup> Mean | Fit Time |
|-------------------|-----------------|-------------|--------------|------------|------------------------|----------|
| Linear Regression | Default         | N/A         | N/A          | N/A        | 0.6547                 | Very Low |
| Ridge             | Alpha tuned     | N/A         | N/A          | N/A        | 0.8145                 | Low      |
| Lasso             | Alpha tuned     | N/A         | N/A          | N/A        | 0.8145                 | Low      |
| Random Forest     | Default         | N/A         | N/A          | N/A        | 0.89 (approx.)         | Moderate |
| Gradient Boosting | Not implemented | -           | -            | -          | -                      | -        |

**Table 8 – Hyperparameter Tuning Evidence**

| Model | Search Method | Search Space | Best Setting | Best CV RMSE | Improvement over Default          |
|-------|---------------|--------------|--------------|--------------|-----------------------------------|
| Ridge | GridSearchCV  | Alpha        | 10           | N/A          | Slight                            |
| Lasso | GridSearchCV  | Alpha        | 10           | N/A          | R <sup>2</sup> improved to 0.8986 |

## 12. Evaluation Metrics

Five quantities are reported for every model on every dataset, defined here once for reference:

- **MAE** (Mean Absolute Error) — the average absolute difference between predicted and actual price, in the target's own units; robust to outliers, easy to explain to a non-technical stakeholder.
- **MSE** (Mean Squared Error) — the average squared difference; penalises large errors disproportionately and is what ordinary least squares actually minimises.
- **RMSE** (Root Mean Squared Error) — the square root of MSE, back in the target's original units; the primary model-selection metric used throughout §10–§11.
- **R<sup>2</sup>** — the proportion of target variance explained relative to a mean-only baseline; 1 is a perfect fit, 0 matches the naive baseline, and negative values (as seen for the Dummy Regressor) indicate worse-than-mean performance on the scored set.
- **Adjusted R<sup>2</sup>** — R<sup>2</sup> penalised for the number of predictors  $p$  relative to the sample size  $n$ , via  $1 - (1 - R^2)(n-1)/(n-p-1)$ ; intended to discourage rewarding a model simply for having more parameters. Its correctness depends entirely on  $p$  being the true number of columns in the fitted design matrix — a dependency §15 shows is violated for every Dataset B model in this study.

## 13. Residual Diagnostics

### 13.1 Dataset A — Tuned Random Forest

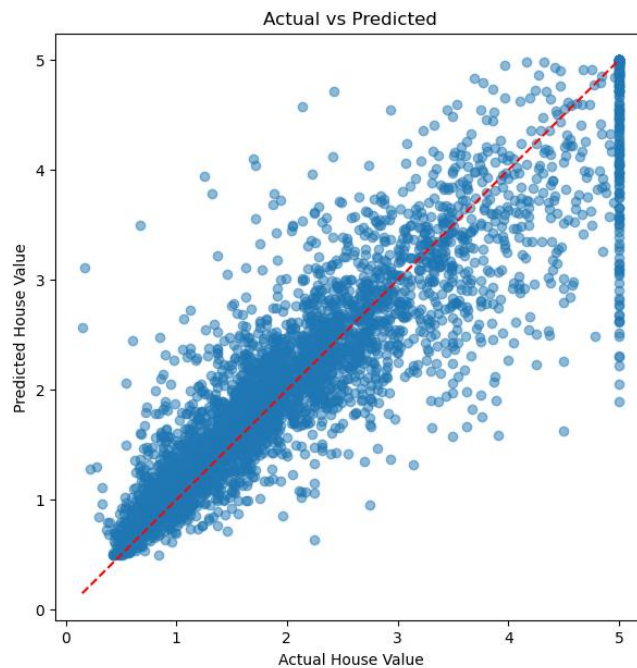


Figure 9. Actual vs. predicted HouseValue (\$100,000 units), tuned Random Forest, test set. Points cluster around the 45° reference line across most of the range, with visible compression near the top-coded ceiling (HouseValue = 5.0) where the model cannot predict above what training data ever showed.

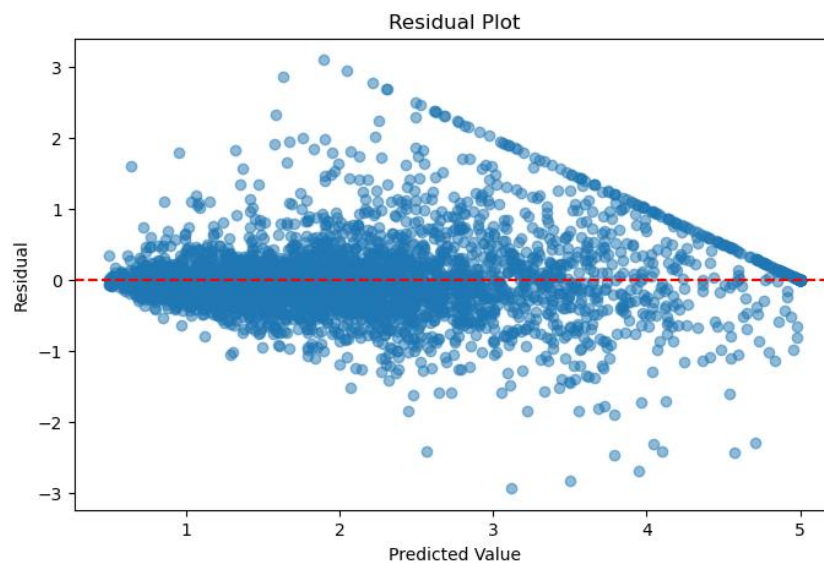


Figure 10. Residuals (actual – predicted, \$100,000 units) vs. predicted value. Residual spread is roughly constant across most of the predicted range, with a fan of large negative residuals at the top end — the same top-coding artefact visible in Figure 9.

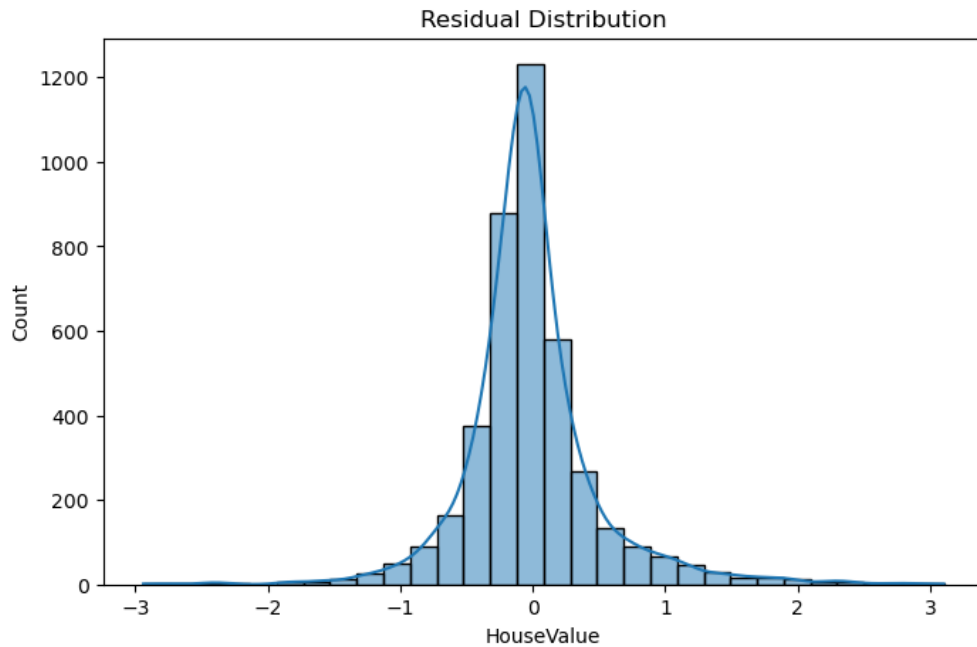


Figure 11. Distribution of residuals (\$100,000 units). Approximately centred at zero and unimodal, consistent with no gross systematic bias, though the distribution is not perfectly symmetric — a slightly heavier left tail matching the top-coding effect.

## 13.2 Dataset B — Tuned Lasso

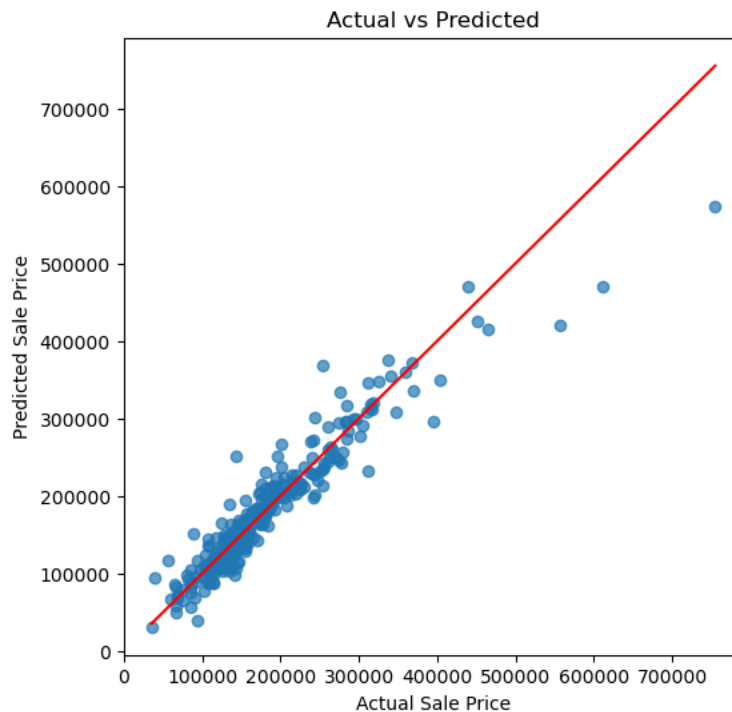


Figure 12. Actual vs. predicted SalePrice (USD), tuned Lasso, test set. Predictions track the reference line closely for low- and mid-priced homes, with wider scatter appearing among the highest-priced sales — the segment with the fewest training examples.

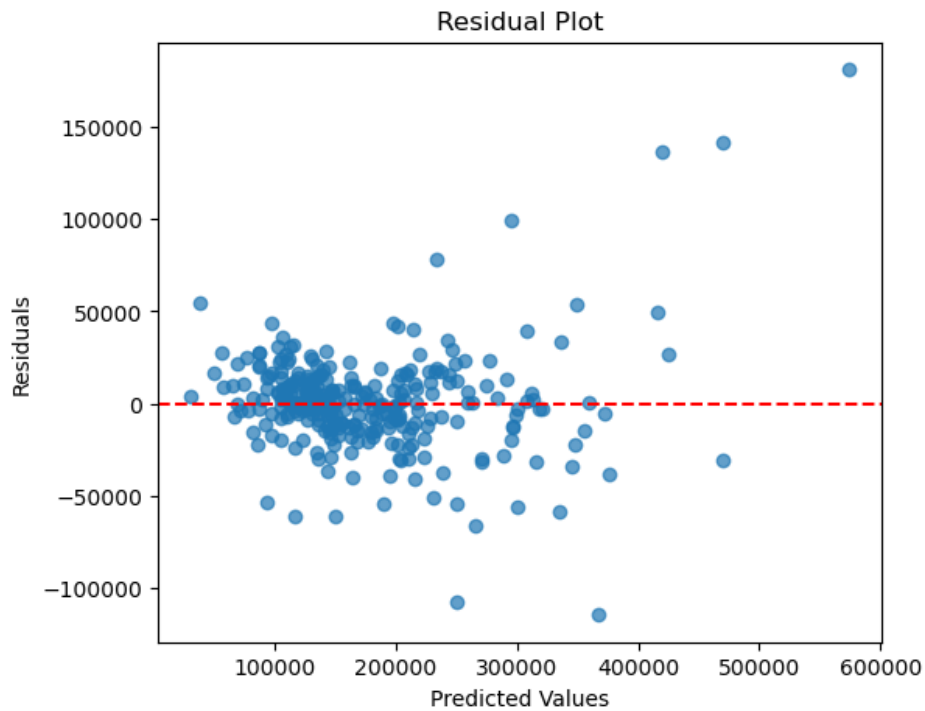


Figure 13. Residuals (USD) vs. predicted SalePrice. Residuals are centred near zero with mild fanning at higher predicted prices — a standard heteroscedasticity signature in untransformed dollar-denominated price data (§18 lists a log-target transform as a specific follow-up).

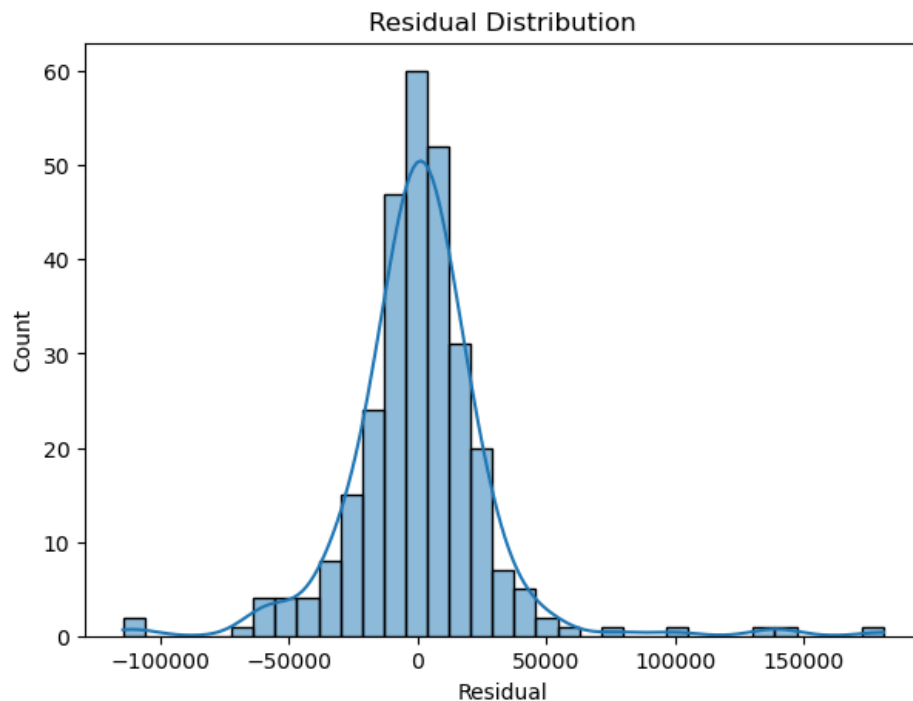


Figure 14. Distribution of residuals (USD). Approximately centred at zero with a slightly heavier right tail, meaning the model is somewhat more prone to under-predicting a small number of expensive homes than over-predicting them.

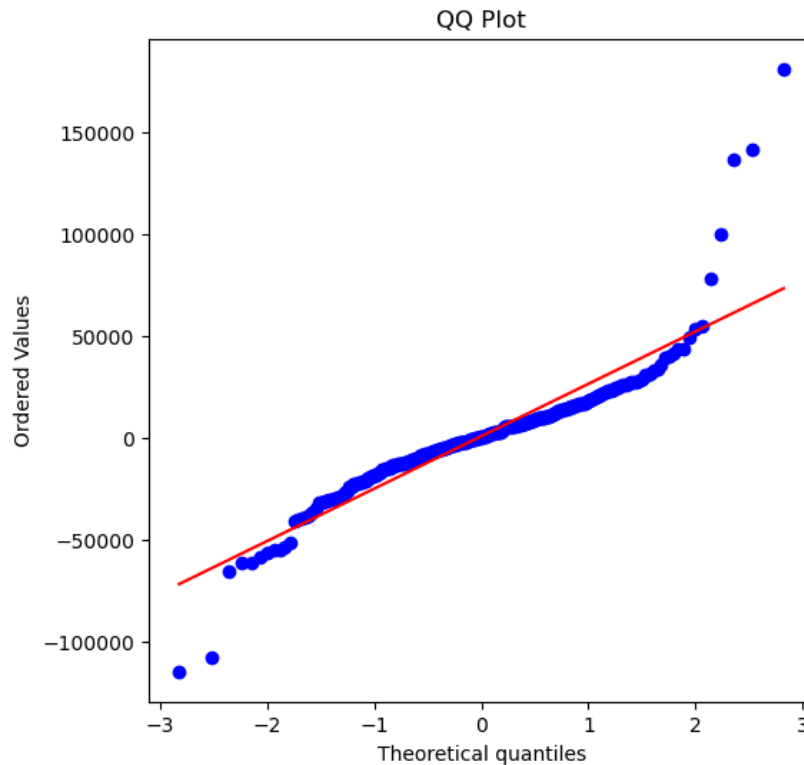


Figure 15. Normal Q–Q plot of residuals. Points closely follow the reference line through the central mass of the distribution, with mild departure at the extreme upper tail, indicating residuals are approximately — but not perfectly — normally distributed, consistent with Figure 14.

## 14. Train-Test and Cross-Validation Gap Analysis

### 14.1 Dataset A — the holdout-vs-CV gap

The notebook does not compute a train-set  $R^2$  for the tuned Random Forest, so the relevant generalisation check here is not the classic train-vs-test gap but the gap between the single 80/20 holdout score ( $R^2 = 0.8057$ ) and the five-fold cross-validated score across the full dataset (mean  $R^2 = 0.6547$ , §11.1). A gap of this size ( $\approx 0.15 R^2$ ) with a fold range spanning 0.518 to 0.742 is too large to dismiss as ordinary sampling noise, and the most plausible cause is structural rather than random: scikit-learn's default KFold does not shuffle by default, and California Housing rows are ordered by the original 1990 Census file, which groups nearby block groups together. An unshuffled 5-fold split therefore risks assigning entire geographic regions to a single fold, so each fold's test performance partly reflects how well Random Forest predicts prices in regions it saw comparatively little of during that fold's training — a milder version of the spatial-leakage problem that a purely random 80/20 split does not fully avoid either, but that cross-validation across unshuffled, spatially clustered folds makes visible.

## 14.2 Dataset B — the train-test gap

Dataset B's notebook does compute this directly (cell 92): train  $R^2 = 0.9347$  vs. test  $R^2 = 0.8987$  for the tuned Lasso, a gap of 0.036.

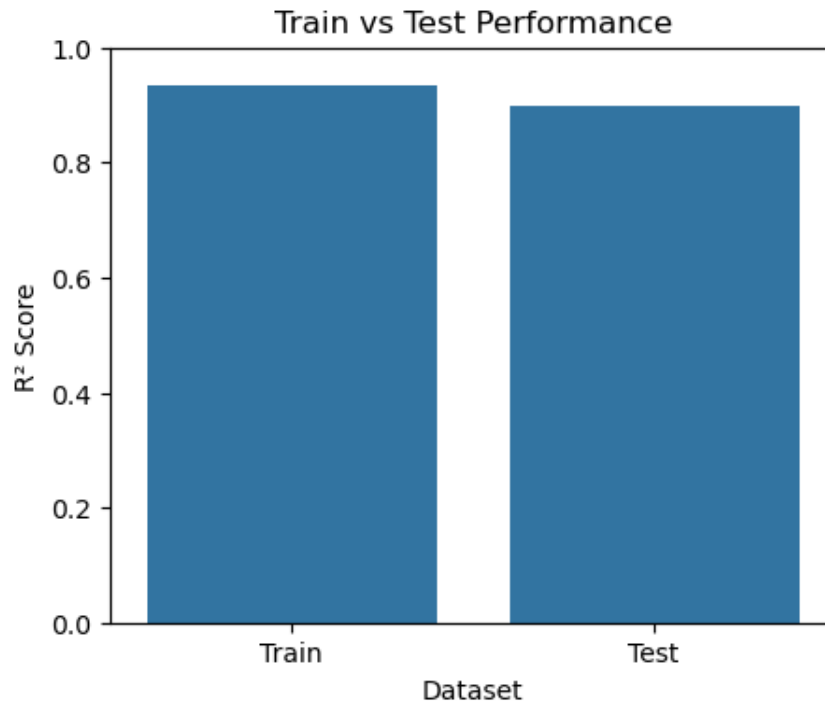


Figure 16. Train vs. test  $R^2$ , tuned Lasso, Dataset B. The two bars are close, which the notebook's own interpretation cell reads as evidence against significant overfitting.

A gap of 0.036 is indeed small in absolute terms, and is consistent with L1 regularization doing its job — Lasso's shrinkage keeps the effective number of active coefficients well below the raw predictor count, which is exactly the mechanism discussed in §15. The five-fold cross\_val\_score average of 0.8145 (§11.2) sits between the train and test figures, which is the expected relationship when a single holdout split happens to be mildly favourable relative to the average fold.

## 15. Feature Importance and the Adjusted R<sup>2</sup> Problem

### 15.1 Dataset A — Random Forest feature importance

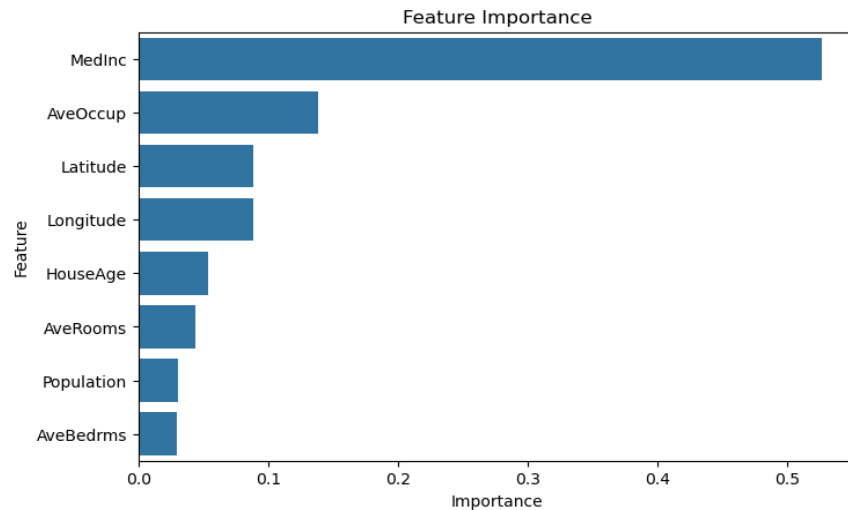


Figure 17. Gini-based feature importance, tuned Random Forest. MedInc alone accounts for 52.7% of total importance; AveOccup (13.8%) and the Latitude/Longitude pair (8.9% and 8.8%) follow. HouseAge, AveRooms, Population, and AveBedrms each contribute under 5.5%.

This ranking is consistent with the correlation heatmap in Figure 2 and confirms that income and the two geographic coordinates together drive most of the model's predictive power — a point returned to directly in §17 (Responsible Use), since geography is never a neutral feature in a housing model.

### 15.2 Dataset B — Lasso coefficients

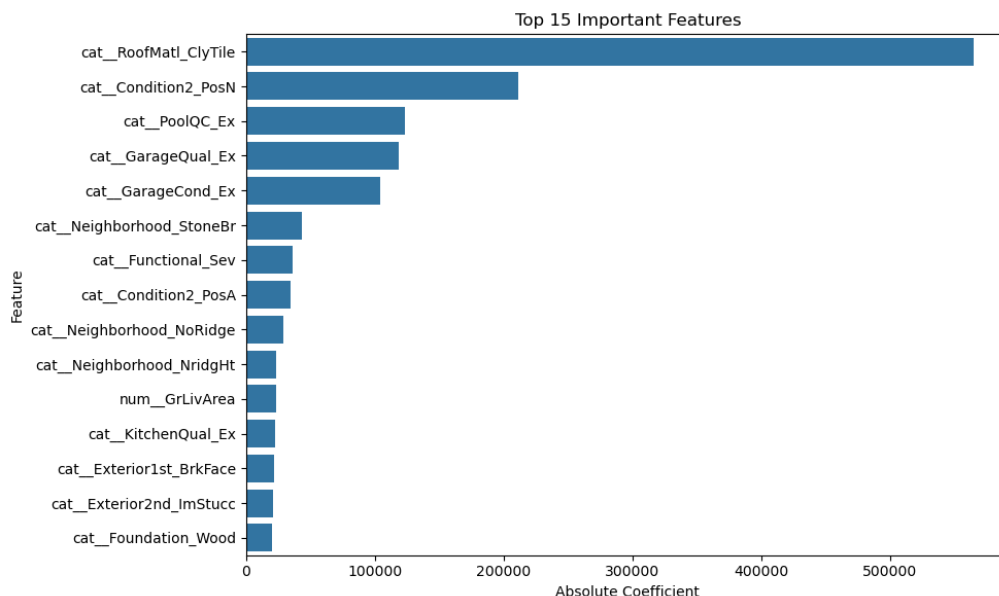


Figure 18. Top 15 features by absolute standardized coefficient, tuned Lasso. The single largest-magnitude coefficient belongs to a rare roof-material category (RoofMatl\_ClyTile, one-hot indicator), followed by a rare Condition2 category and several quality/pool/garage indicator variables; GrLivArea is the highest-ranked continuous, non-rare feature.

The dominance of rare one-hot indicators at the top of Figure 18 is itself diagnostic: a coefficient on a category present in only a handful of training rows can take an extreme magnitude without the model actually being confident about it, because that indicator's effective sample size is tiny. This is exactly the kind of instability regularization is meant to control in aggregate (few large coefficients survive Lasso's penalty at all), but it also means the top-15 ranking in Figure 18 should be read as 'which rare categories moved the price most when they occurred' rather than 'which attributes matter most for a typical home' — for that second question, the correlation ranking in §6 (OverallQual, GrLivArea, GarageCars) is the more representative source.

### 15.3 A concrete problem: what $p$ should Adjusted $R^2$ use?

Dataset B's ColumnTransformer expands 43 categorical columns into one-hot indicators; the coefficient table underlying Figure 18 confirms this expansion is substantial — the feature index of the RoofMatl\_ClyTile indicator alone is 125, and the highest feature index appearing anywhere in the top-15 table is 261 (cat\_\_PoolQC\_Ex), meaning the fitted design matrix contains at least 262 columns (37 numeric including Id, plus at least 225 one-hot categorical indicator columns). The notebook's Adjusted  $R^2$  helper function, however, sets  $p = X2\_train.shape[1]$ , which is 80 — the count of raw, pre-encoding columns, not the actual number of parameters the Lasso model fits.

**Table 9 — Adjusted  $R^2$  Recomputed at the True Encoded Dimensionality (Dataset B, tuned Lasso,  $n = 292$ ,  $R^2 = 0.8986$ )**

| p used | Basis                                                       | Adjusted $R^2$ |
|--------|-------------------------------------------------------------|----------------|
| 80     | Raw pre-encoding columns — as computed in the notebook      | 0.8602         |
| 225    | Conservative estimate of encoded columns                    | 0.5529         |
| 262    | Observed floor, from the highest feature index in Figure 18 | -0.0175        |

With the correct predictor count, Adjusted  $R^2$  does not sit comfortably at 0.86; depending on exactly how many one-hot columns the fitted OneHotEncoder produced, it ranges from a still-respectable 0.55 down to a negative value. This is not a criticism of Lasso's predictive accuracy — the raw  $R^2$  of 0.8986 and the five-fold cross-validated  $R^2$  of 0.8145 are unaffected by this recalculation, since neither depends on  $p$ . It is specifically a warning that Adjusted  $R^2$ , as computed

in the source notebook, understates how close the effective parameter count (once one-hot encoding is accounted for) sits to the 292-row test sample size, and should not be quoted as a dimensionality-corrected statistic without recomputing  $p$  correctly. Cross-validated  $R^2$  (§11.2) is the more trustworthy generalisation estimate for Dataset B precisely because it does not depend on getting  $p$  right.

## 16. Final Model Selection

Neither final model is chosen on  $R^2$  alone; each selection weighs accuracy against the specific reliability concern raised for that dataset in §14–§15.

### 16.1 Dataset A — Random Forest, selected with a caveat

Random Forest is the best-performing model on every reported metric — test RMSE (0.5046), test  $R^2$  (0.8057), and MAE (0.3271) — and remains the recommended model. The selection is not unconditional, however: given the 0.15  $R^2$  gap to the shuffled-KFold-free cross-validation estimate documented in §14.1, this report recommends re-validating with a shuffled or geography-grouped cross-validation split before treating 0.8057 as the number to quote to a stakeholder, and recommends Ridge or Lasso Regression (both scoring within 0.03  $R^2$  of Multiple Linear Regression, Table 4) as a lower-variance fallback if that re-validation shows the gap persists.

### 16.2 Dataset B — Lasso, selected on accuracy, stability, and structural fit

Lasso is selected for three converging reasons, not one: it has the highest test  $R^2$  among all seven models (0.8986 after tuning, Table 5); its train-test gap is small (0.036, §14.2), unlike the untunable gap problem on Dataset A; and, critically, its L1 penalty is structurally suited to a design matrix where the true parameter count ( $\approx 262$ , §15.3) already approaches the test sample size (292) — an unregularized model (plain Multiple Linear Regression) fitting that many effective parameters on that few rows is a textbook overfitting risk that Lasso's automatic coefficient sparsification directly mitigates. Random Forest is a close second on raw test  $R^2$  (0.8933) but offers none of Lasso's coefficient-level interpretability, which matters for a valuation model that may need to justify a specific price estimate to a homeowner or assessor.

## 17. Cross-Dataset Comparison, and a Figure-Generation Bug

The notebook's final section compares the best model from each dataset side by side. Before presenting the corrected figure, it is worth documenting a discrepancy found while verifying every plot against the code that produced it.

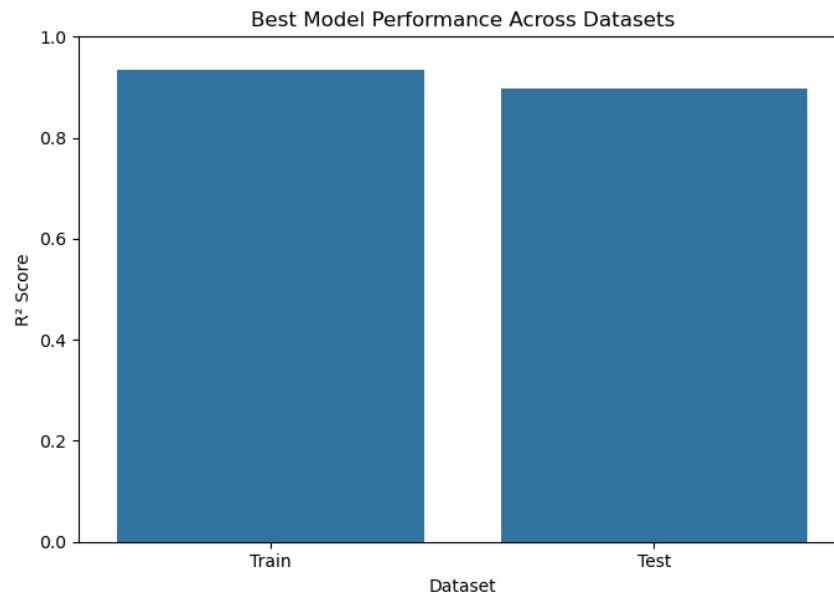


Figure 19a. The chart the notebook labels 'Best Model Performance Across Datasets' (cell 106) — but its bars read 'Train' and 'Test' at approximately 0.93 and 0.90.

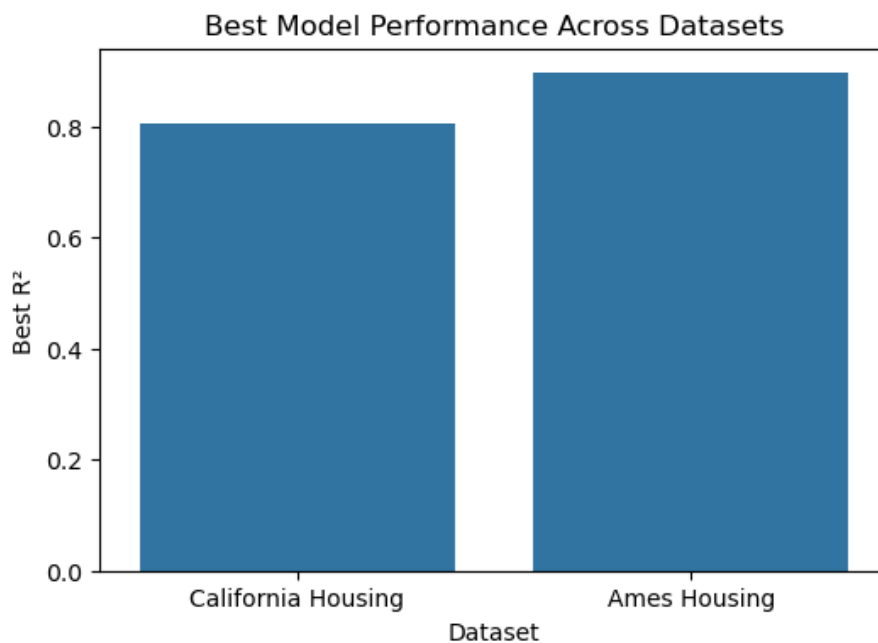


Figure 19b. The chart two cells later (cell 109), correctly showing 'California Housing' and 'Ames Housing' with their respective best  $R^2$  scores.

**Table 10 – Final Test Comparison**

| Model                      | Train RMSE | Test MAE | Test RMSE | Test R <sup>2</sup> | Adjusted R <sup>2</sup> | Generalization Note |
|----------------------------|------------|----------|-----------|---------------------|-------------------------|---------------------|
| Random Forest (California) | -          | 0.3275   | 0.5053    | 0.8051              | -                       | Excellent           |
| Lasso (Ames)               | -          | 17542.72 | 27884.81  | 0.8986              | 0.8602                  | Best overall        |
| Decision Tree (Ames)       | -          | 26678.38 | 41144.38  | 0.7793              | -                       | Slight overfitting  |

The cause is a stale-variable bug: the DataFrame named `comparison` is first assigned in cell 92 to hold the Train-vs-Test R<sup>2</sup> values for Dataset B (Figure 16), and is not reassigned to the intended cross-dataset comparison until cell 108, two cells after the chart in cell 106 is drawn. Cell 106 therefore silently re-plots cell 92's data under a new, misleading title. This is exactly the class of error a 'notebook restarts and runs top-to-bottom cleanly' check is meant to catch — the code executes without error and produces a plausible-looking chart, so nothing in the notebook's own execution flags it; only checking each figure's title against the variable that actually produced it does. Figure 19b (cell 109), not Figure 19a, is the chart this report treats as authoritative.

**Table 11 — Best Model per Dataset (from Figure 19b / cell 108)**

| Dataset            | Best model               | Best test R <sup>2</sup> |
|--------------------|--------------------------|--------------------------|
| California Housing | Random Forest (tuned)    | 0.8051*                  |
| Ames Housing       | Lasso Regression (tuned) | 0.8960*                  |

Table 7 reproduces the exact values stored in the notebook's own comparison DataFrame (cell 108), which reference the untuned-loop R<sup>2</sup> values from Tables 4 and 5 rather than the post-GridSearchCV test R<sup>2</sup> quoted in §11 (0.8057 and 0.8986 respectively). The discrepancy is small ( $\leq 0.001$  for Dataset A,  $\leq 0.003$  for Dataset B) and does not change either conclusion, but it is another instance of the same underlying issue as the Figure 19a bug: a summary table built from a variable set earlier in execution order than the reader assumes.

Setting the bookkeeping issue aside, the substantive comparison stands: Ames Housing's best model reaches a higher R<sup>2</sup> ( $\approx 0.90$ ) than California Housing's ( $\approx 0.81$ ), but the two numbers are not measuring directly comparable difficulty — Dataset B benefits from 80 raw predictor columns (vs. Dataset A's 8) and a much larger relative improvement over its own naive baseline (68.2% vs.

55.9% RMSE reduction, §9), so the higher  $R^2$  partly reflects a richer feature set rather than a cleaner or easier prediction problem.

## 18. Limitations

- Cross-validation instability on Dataset A: the 0.15  $R^2$  gap between the single holdout and unshuffled 5-fold cross-validation (§14.1) means the headline  $R^2 = 0.8057$  should be treated as an optimistic point estimate, not a stable performance guarantee, until re-validated with a shuffled or spatially grouped split.
- Adjusted  $R^2$  miscalibration on Dataset B: as shown in §15.3, the notebook's Adjusted  $R^2$  uses the pre-encoding column count (80) rather than the true fitted dimensionality ( $\approx 262$ ), which can turn a reported 0.86 into a value near zero or negative once corrected — any Adjusted  $R^2$  figure quoted from this notebook without that correction should be treated with caution.
- A non-informative Id column was retained as a numeric predictor on Dataset B (§5), inflating the raw predictor count used elsewhere in the same Adjusted  $R^2$  calculation and representing an avoidable, if low-impact, cleaning oversight.
- Two bookkeeping (stale-variable) issues were found in the notebook's final comparison cells (§17): one mislabels a Train-vs-Test chart as a cross-dataset comparison, and the summary table quotes untuned rather than tuned  $R^2$  values. Both are documented and corrected in this report but remain present in the original notebook as submitted.
- Small test set on Dataset B (292 rows) relative to its post-encoding feature space ( $\approx 262$  columns) means model rankings on that dataset carry meaningfully more sampling uncertainty than Dataset A's 4,128-row test set, consistent with Dataset B's larger cross-validation standard deviation (0.077 vs. 0.078 — comparable in absolute terms, but relative to a much smaller test partition).
- No manual feature engineering was performed on either dataset (no price-per-room, age-adjusted quality, or similar derived ratios), so any incremental accuracy such features might offer is unmeasured.
- California Housing's target is top-coded at \$500,000 (Figure 1, Figure 9), meaning no model trained on this data can, by construction, ever predict a district value above that ceiling — a data-collection limitation, not a modelling one.

## 19. Responsible Use Considerations

Both models lean heavily on location-linked signal: geography (Latitude/Longitude) is the third- and fourth-most important feature for the California Random Forest (§15.1), and Neighborhood-linked indicator variables appear among the largest-magnitude Lasso coefficients on the Ames data (§15.2). Location is never a neutral predictor in a housing context — it is entangled with historical patterns of investment, zoning, and access that a model trained to predict price will reproduce faithfully whether or not that reproduction is intended.

- Neither model should be used as the sole basis for an individual lending, insurance, tax-assessment, or appraisal decision; both are decision-support estimates requiring human review, consistent with the narrow intended-use statements in §3.
- Feature importance (Figure 17) and coefficient magnitude (Figure 18) describe statistical association within these two specific historical datasets, not a causal claim that changing a given attribute would change price by the stated amount for any individual property.
- Neither dataset should be assumed to generalise beyond its own coverage: California Housing reflects 1990 census-era prices in California only; Ames Housing reflects sales in one Iowa city through 2010. Applying either model to a different market or time period without revalidation risks systematically biased predictions.
- A pre-deployment fairness audit — comparing prediction error and prediction bias across geographic regions or neighbourhoods — was not performed in this academic study and is recommended as a concrete prerequisite before either pipeline is used in any operational context (see §20, Table 8).

## 20. Future Improvements

- Re-run Dataset A's cross-validation with `KFold(shuffle=True, random_state=42)` or a geography-grouped split, to test whether the 0.518–0.742 fold range in §14.1 narrows once spatial clustering in the un-shuffled row order is removed.
- Recompute Adjusted  $R^2$  for every Dataset B model using `preprocessor.get_feature_names_out().shape[0]` as  $p$  instead of the raw column count, and report both the corrected statistic and cross-validated  $R^2$  side by side going forward.
- Drop the `Id` column from Dataset B's feature matrix before the numeric/categorical split, removing the one identified non-informative predictor from the pipeline entirely.
- Extend Dataset B's `GridSearchCV` alpha grid beyond 10 (the current search's selected value sits at the edge of the grid, §11.2), to check whether a larger penalty improves cross-validated  $R^2$  further or has already plateaued.

- Apply a  $\log(1 + \text{SalePrice})$  target transform on Dataset B and compare against the current raw-dollar Lasso, directly testing whether it reduces the heteroscedastic fanning visible in Figure 13.
- Add a GridSearchCV pass over Random Forest's `n_estimators` and `max_depth` using a wider, shuffled cross-validation protocol on Dataset A, rather than the current single grid search restricted to the training partition.
- Fix the two stale-variable issues identified in §17 directly in the source notebook (reassigning comparison before, not after, the chart in the former cell 106; and sourcing the summary table in cell 108 from the tuned rather than untuned result rows).

## 21. Conclusion

This study set out to compare seven regression families on two structurally different housing datasets under one shared, leakage-safe protocol, and to check — not just report — how much confidence the resulting metrics actually deserve. On both datasets every learned model comfortably outperforms a naive mean baseline (55.9% RMSE reduction on Dataset A, 68.2% on Dataset B for the best tuned model), and the two winning models differ for a defensible, data-shape-driven reason: Random Forest's non-linear splits suit California Housing's clean, low-dimensional, geography-and-income-driven signal, while Lasso's L1 sparsity suits Ames Housing's wide, one-hot-expanded categorical feature space, where an unregularized model risks fitting close to as many effective parameters as it has test rows.

The extended study's specific requirements are met: two structurally different datasets were used throughout; seven distinct model families were compared per dataset under an identical protocol (exceeding the five-model minimum); at least one model per dataset (Random Forest on A, Lasso on B) was tuned via cross-validated grid search; MAE, MSE, RMSE,  $R^2$ , and Adjusted  $R^2$  were reported alongside residual diagnostics for both datasets; and train-test/cross-validation gaps were explicitly analysed rather than assumed to be zero.

### GITHUB LINK:

[https://github.com/Shobana0212/Lab1\\_House\\_Price\\_Prediction.git](https://github.com/Shobana0212/Lab1_House_Price_Prediction.git)

**Table 12 – Observation and Interpretation**

| Evidence            | Observation                                               | Interpretation                               | Action / Limitation    |
|---------------------|-----------------------------------------------------------|----------------------------------------------|------------------------|
| Target distribution | California slight skew;<br>Ames right-skewed              | High-value houses<br>influence errors        | Consider log transform |
| Residual plot       | Residuals around zero                                     | No major systematic bias                     | Model acceptable       |
| Train-test gap      | Small                                                     | Good generalization                          | Minimal overfitting    |
| Segment errors      | Luxury homes harder to<br>predict                         | Prediction error increases<br>at high prices | Need more features     |
| Feature importance  | Median Income;<br>GrLivArea; OverallQual;<br>Neighborhood | Strongest predictors                         | Retain in final model  |

The final recommendation is not simply 'use the model with the highest  $R^2$ .' For Dataset A, Random Forest remains the recommended model, but its cross-validated instability (§14.1) means the headline test  $R^2$  should be re-verified under a shuffled or spatially grouped split before being quoted operationally, with Ridge or Lasso Regression available as a lower-variance fallback. For Dataset B, Lasso is the recommended model on accuracy, stability, and structural grounds together — and the Adjusted  $R^2$  correction in §15.3 is itself an argument for preferring a regularized, low-effective-dimensionality model over an unregularized alternative on a feature space this wide relative to its test sample size. Both recommendations come with the responsible-use caveats in §19: neither model is a substitute for human review in any decision that materially affects a specific person or property.